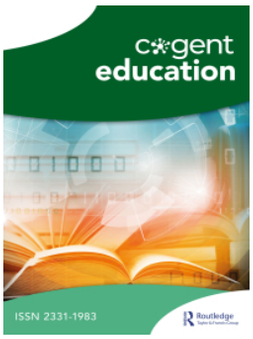

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ChatGPT use in universities in South Africa through a socio-technical lens

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ABSTRACT

As artificial intelligence proliferates, so do associated hopes and fears. This study explores such tensions within South African higher education following ChatGPT's launch, analyzing perceived threats alongside opportunities for responsibly harnessing benefits. Adopting a socio-technical framework recognizing technology's interdependence with social systems, it investigates institutional uncertainties regarding unchecked AI adoption. Whilst findings reveal significant concerns about impartiality, critical faculties and employment impacts, evidence also indicates artificial intelligence, if transparently governed, could enrich scholarship. However, effectively leveraging such opportunities necessitates updated policies fostering accountable assimilation rather than reactionary resistance. Through a systematic literature analysis, this study contends that conceptualizing tools like ChatGPT as amplifying rather than automating academia's technical capacities, with protocols ensuring human oversight, provides the most constructive paradigm. Rather than technologies threatening academics' relevance, an agile, ethical integration strategy upholding rigorous pedagogical, research and assessment standards while expanding inclusion and insight is advocated. The conclusions caution against vilifying innovation but urge policymakers to foster frameworks transparently aligning artificial intelligence's trajectory with academic integrity. Ultimately, the socio-technical approach promoted underscores technological disruption's potential, if governed accountably, to strengthen rather than undermine scholarly excellence.

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Introduction

Artificial intelligence has developed rapidly in recent decades and has had a significant impact on academia (Zawacki-Richter et al., 2019). Following the launch of ChatGPT in November 2022, uncertainties and fears have emerged within the academic community, especially in the global South. Drawing on socio-technical systems theory (Trist, 1981; Griffith & Dougherty, 2001), this paper explores the response of South African academics to ChatGPT. The theory contends that organizations comprise interacting social and technical subsystems that must align for the system to function effectively (Pasmore et al., 1982). The technical subsystem encompasses technologies like ChatGPT, whilst the social subsystem includes the professors who may use such tools. Introducing new technology like AI into universities expands the technical subsystem. However, the theory posits that rather than being a threat, technology should play a complementary role – the two subsystems must align. This paper argues that instead of dismissing ChatGPT as jeopardizing academic integrity, it should be viewed as augmenting the technical capacities of universities. However, policies and protocols are needed to foster responsible alignment between the two subsystems.

The study utilizes a socio-technical lens to interrogate uncertainties expressed by South African academics towards the unregulated use of ChatGPT in higher education. The key contention is that by viewing

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ChatGPT as an extension of the technical subsystem, appropriate policies can ensure it complements rather than replaces the social (human) subsystem. This entails professors utilizing AI tools responsibly whilst retaining critical thinking and analysis. The study recommends constructing policies, informed by strides in other sectors, to align technical innovations like ChatGPT with upholding rigorous academic standards.

ChatGPT

As an AI system capable of natural language processing, ChatGPT represents a noteworthy expansion of the technical capacities available to academia. Developed by OpenAI and underpinned by generative preprocessing and deep learning, ChatGPT can parse textual data, identify linguistic relationships, and generate coherent, contextually relevant responses (Liang et al., 2022). OpenAI announced,

> We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer follow-up questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests (OpenAI, 2023).

This enables nuanced dialogue, question answering, and content creation – augmenting human capabilities regarding analysing and producing written material. However, whilst representing impressive technical progress, systems like ChatGPT also introduce uncertainties regarding impacts on human roles and oversight. As this study's introduction outlined, socio-technical systems theory contends that rather than technology replacing people, an equilibrium must exist between the social and technical subsystems for organizational functionality (Trist, 1981). Introducing disruptive innovations like AI expands the technical subsystem, requiring corresponding recalibration of policies, training, and assessment to ensure ethical, accountable usage that complements without substituting human strengths.

Specifically, within academia, unchecked adoption of conversational systems like ChatGPT risks undermining critical analysis, integrity, and rigor whilst threatening jobs involving information provision. However, appropriate incorporation following socio-technical principles of complementarity could enrich pedagogy, research, and assessment. Indeed, Zawacki-Richter et al. (2019) propose an AI-supported methodology enhancing scholarly productivity using writing aids. This entails transparently applying tools like ChatGPT in restricted, defined contexts to summarize literature and generate essay outlines for students to build upon using their reasoning, ideological perspectives, and evidence integration skills.

In essence, whilst representing an exponential growth in technical capacities, systems like ChatGPT underscore the urgency of developing policies, training programs and pedagogical reforms centred on ethical AI alignment, upholding academic standards whilst harnessing benefits. Rather than resist or accelerate adoption, a considered, socially-attuned strategy for assimilation is recommended – upholding human strengths while responsibly leveraging AI's pattern recognition and language proficiency to enrich scholarship and learning.

Theoretical framework

This paper draws on the socio-technical system theory. This theory proposes that technology is not just a tool but also a social system and that the interaction between technology and the social system is what creates its impact (Greenhalgh, 2013). The socio-technical system theory is an interdisciplinary approach that examines the complex interplay between social and technical elements of a system. The theory posits that there are two main types of systems: technical systems and social systems. Mumford (2006) aver that technical systems are comprised of the physical and technological components of an organization, such as machines, tools, and computer systems. Social systems, on the other hand, are made up of people, their roles, and the relationships they have with each other. The theory emerged in the 1950s and 1960s as a response to the increasing use of technology in the workplace and the recognition that technical systems were not operating as effectively as they could because of the social factors that surrounded them (Emery & Trist, 1960; Mumford, 2006). The development of the socio-technical system theory was influenced by several key thinkers, including Eric Trist, A.K. Rice, and Fred Emery. These scholars argued that organisations were made up of social and technical subsystems that were interdependent and that both subsystems needed to be aligned for the organisation to function

effectively. According to Pasmore et al. (1982, p. 1182), the socio-technical theory is, 'method of viewing organisations which emphasises the interrelatedness of the functioning of the social and technological subsystems of the organisation and the relation of the organisation to the environment in which it operates. Put simply, the sociotechnical system perspective contends that organisations are made up of people that produce products or services using some technology, and that each affects the operation and appropriateness of the technology as well as the actions of the people who operate it'. Embedded at the epicentre of this definition is the important notion of value addition, whereby the products and services produced are 'valued by customers (who are part of the organisation's external environment)' (Griffith & Dougherty, 2001, p. 206). Furthermore, the theory argues that human input should not be relegated to the periphery when new technology is introduced into the workplace. In this paper, AI is viewed as part of the technical system while academics using AI are the social system. Thus, the integration of ChatGPT has to be premised on the understanding that usage of ChatGPT must not entail outsourcing thinking to the technical system. However, components within a system must be complementary and therefore ChatGPT must be viewed as complementary and not a replacement for academics. Furthermore, drawing from the socio-technical theory it can be argued that ChatGPT (technical system) and the academics (social system) must be aligned for universities to function effectively.

Furthermore, ChatGPT integration in universities can also be viewed as expanding the technical system. The socio-technical system theory also emphasizes the importance of considering the broader context in which the technology is being used (Emery & Trist, 1960; Mumford, 2006; Trist, 1981). In the case of ChatGPT in academia, this would involve considering factors such as the institutional culture, the goals of the educational program, and the ethical implications of using AI in education.

Research methodology

Guided by calls to detail the literature review process, this study adopted systematic review principles entailing a phased search strategy with inclusion/exclusion criteria to enhance replicability (Gough et al., 2017). Aligning with the socio-technical systems framework emphasizing ethical AI assimilation, the review sought to collate perspectives, trends and debates regarding ChatGPT's application in higher education to inform policy and practice.

The literature search encompassed multiple academic databases including Google Scholar, Scopus, Web of Science and ERIC to enable capturing diverse insights across disciplines including education, technology, ethics, and society. Search terms included 'ChatGPT', 'AI', and 'higher education'. Peer-reviewed articles were prioritized, but edited books, conference papers, reports and media articles provided contextual insights and future projections. However, a review of articles on social media, news articles and blogging platforms such as Twitter and Medium Daily Digest yielded salient additional responses, given the rapidly evolving public debates regarding ChatGPT often preceding academic literature cycles. Whilst these non-peer-reviewed perspectives were more interpretative, they provided vital context regarding concerns like threats to assessment integrity and faculty job automation that peer-reviewed scholarship is still crystallizing. Publications exclusively focused on school-level education were excluded given the study's higher education scope.

Regarding date parameters, whilst findings were collated until data analysis completion in April 2023, no commencement date limit was set since discussions regarding automation, AI and higher education have intensified over recent decades. However, 87% of included sources were published from 2020 onwards, capturing burgeoning ChatGPT-specific analysis. Thematic analysis was then conducted, coding articles by cross-cutting themes pertaining to opportunities, risks, policies, and recommendations regarding AI-adoption in universities to address the study's objectives. Literature was specifically analysed regarding balancing ChatGPT's integration with upholding academic integrity per the socio-technical framework's tenets.

The multi-step search strategy, inclusion of diverse literature and phase-based theming thereby enabled a rich, contextualized analysis – providing a springboard to advance policy and practice discussions regarding ethically harnessing AI's opportunities whilst mitigating its risks. The methodology also allowed targeting literature intersections, for example, spanning technical descriptions of ChatGPT alongside questions of accessibility, automation anxieties and academic integrity. Such multifaceted insight

supports developing a balanced, evidence-based standpoint on AI assimilation – upholding standards whilst responsibly expanding technical capabilities.

Arguments on the use of ChatGPT in universities

This section presents various arguments on the use of ChatGPT in the academia. The arguments are presented using themes. The following themes guide this discussion: ChatGPT as a threat to academia and academic integrity and potential benefits of using ChatGPT.

ChatGPT as a threat to academia and academic integrity

Throughout history, responses to innovation have been varied, often influenced by one's perspective. In academia, initial reactions to ChatGPT have predominantly been concerned with perceiving it as a threat. This perception is based on ChatGPT's demonstrated capabilities. Mushtaq (2023) notes that ChatGPT has made academics anxious all over the world. This surprising reluctance to embrace ChatGPT by academics can be viewed as contradictory to the norm that academia are trendsetters and leaders in adopting innovations. The reluctance was informed by the view that if ChatGPT can be utilised to answer questions, then it can certainly be used by students to do their assessments. Apocalyptic arguments are based on the view that students will no longer be motivated to research for their assignment as they will be relying on ChatGPT. These fears are not unfounded as essays and assignments have been at the epicentre of tertiary institutions. If AI can now be used to do work for students like write assignments, then academics are justified in getting concerned to a certain extent.

Additionally, another reason some academics are cautious about ChatGPT, is that it can generate highly convincing and sophisticated text, which could potentially be used to produce fake or misleading research articles or other types of content. This could undermine the integrity of academic research and make it more difficult to distinguish between real and fake information. Misleading research articles negatively impact the learning of students as some students may not be able to discern fake articles from authentic peer-reviewed research. Additionally, the other argument put forth by those who view ChatGPT as a threat is rooted in the concern that it may eventually replace human academics in certain areas. This worry is based on the impressive ability of ChatGPT to generate human-like responses. Another concern is that the use of AI models like ChatGPT could lead to a reduction in critical thinking in academic writing. If researchers become overly reliant on ChatGPT to generate text or ideas, they may lose their ability to think creatively and independently. Thus, if students become also reliant on the use of AI, critical thinking, and originality will thus be adversely impacted according to this school of thought. However, applying a socio-technical approach to the adoption of ChatGPT by students and professors, will entail complementarity and not replacement of thinking. Hence dispelling the view that ChatGPT is a threat to critical thinking. Additionally, these apparent concerns led the author to pose and reflect on the Chinese proverb that says, '*When the winds of change blow, some people build walls and others build windmills*'. According to this old Chinese wisdom, whenever there is impeding change, people either choose to build walls to try and stall the progress while others take advantage of that change. The academia now finds itself in a quandary on whether to build walls or windmills. The red tapes in universities that were designed to protect academic integrity have persuaded some academics to build walls. After all the yearbooks and policies are clear on the need for students to take their assessments without any help and using ChatGPT is translated as cheating. However, surprisingly students as well as academics have been using Grammarly among other AI tools to edit their work, but it has never been considered cheating. The socio-technical theory, however, can be utilised to counter these arguments as ChatGPT should be viewed as an addition to the technical system already in use in universities and therefore can be utilised in a complementary way. All that is needed is to align the social system, that is the students and professors and the technical system to ensure efficacy in universities.

Additionally, the ChatGPT threat to academia argument can be extended to implications on academic integrity. Building walls appears as a more convenient option for academics because it is safer to err on the side of caution. What if the use of ChatGPT is later officially considered cheating and academics end up losing their hard-earned and protected reputations as scholars? No academic wants to have their

academic integrity brought into question just because they have utilised ChatGPT in doing their work after decades of hard work. Within the South African university space, academics are quickly reminded of the case of an academic who published five articles between 2012 to 2014 in a journal that was later established to be predatory (Maistry, 2019; Le Grange, 2019). Publishing in the predatory journal severely impacted on the academic's integrity. The uncertainty on the ethicalness of the use of ChatGPT informs the approach by some academics to be cautious. Maistry (2019, pp. 5–6) explains, 'academics are expected to be astute intellectuals, inherently critical and constantly suspicious, possessing robust scientific publishing literacy'. Fundamentally, there is a need to interrogate the implications of the use of ChatGPT by academics in their research work on intellectual astuteness and scientific robustness alluded to by Maistry (2019). Further questions have been raised on how to acknowledge the contribution of ChatGPT in research papers. Not acknowledging the contribution of ChatGPT to any academic work poses ethical issues. Fraser (2014) and Park (2003) define plagiarism as 'literary theft'. Plagiarism is an academic offense involving the use of other people's work without acknowledging it. These debates have led to calls to redefine plagiarism in response to the proliferation of AI-generated content (Tarisayi, 2023). Tarisayi (2023) studies university policies in South Africa and argued for a relook at the plagiarism policies. In pursuit of academic integrity, recent articles in the field of medicine have included ChatGPT as an author (Stokel-Walker, 2023). Stokel-Walker (2023) writes 'at least four articles credit the AI tool as a co-author, as publishers scramble to regulate its use'. Though the inclusion of ChatGPT as an author might be an effort to be transparent, questions arise on the basics of authorship as it was previously understood. Some journals require authors of papers to clearly state the contribution of each author listed on a publication. Furthermore, each of the listed authors is also required to sign an author declaration. Therefore, if ChatGPT is included as an author who will sign the author declaration? An author must be held accountable for their contribution to a paper and how is that possible with ChatGPT as an author? At times authors are asked to retract their work as a way of holding them accountable. Thus, leaving academia in a further quandary on whether to list ChatGPT as an author or not.

Furthermore, concerns around the question of how long ChatGPT is going to be available free. ChatGPT was released as a free research preview. Later, Open.ai launched an upgrade, ChatGPT Plus that requires subscriptions. The launching of ChatGPT Plus buttressed the concerns raised by some academics that eventually the service is going to be monetised. In an unequal society like South Africa, requiring subscriptions to access AI inevitable widens the gap between students at better-resourced universities and students at disadvantaged universities. Additionally, these uncertainties were compounded by the recurring ChatGPT outages in early 2023. ChatGPT reportedly had roughly between 15 and 20 incidents in a space of two months in 2023 according to AI blog, AI Whisperer (2023). These outages were attributed to the infrastructure struggling to cope with the millions of users and trillion requests at single point of time. Novice users are afraid of depending on a system that could fail at crucial moments, particularly in academic settings. Academics diligently prepare their work ahead of time and typically test their systems beforehand. It would be disappointing for them if ChatGPT experiences an outage during their planned usage. However, these ChatGPT outages confirm the central argument in the socio-technical theory that humans should not be relegated to the periphery by technology. ChatGPT outages can be utilised to warn users not to surrender all their thinking to AI but use it to complement their work. The next section discusses the likely benefits of using ChatGPT within the academia.

Potential benefits of using ChatGPT

While on the other hand, academics have the alternative to build windmills and take advantage of the advantages that come with using ChatGPT. An emerging voice on the use of AI in academic writing on the social media platform, Twitter, Mushtaq Bilal advocates for using ChatGPT intelligently (and ethically) for academic purposes. Mushtaq (2023) premises his argument with the bold declaration that 'ChatGPT is everywhere, and everyone is using it'. Even though this might be a far-stretched generalisation to say that everyone is using it, it should be noted that the use of ChatGPT is increasing astronomically. Most notably are statistics that reveal that ChatGPT reach one million users within five days. It is interesting to note that it took Facebook 10 months and Twitter two years respectively to reach one million users (Buchholz, 2023). While Dan Milmo writing in the Guardian reports that ChatGPT reached 100 million

users in two months (Milmo, 2023). The uptake of ChatGPT despite being astronomical in the developed world as revealed by the above statistics, it will take more time in the developing world where it is always about catching up on innovations. Aubert (2004, p. 2) notes, 'Innovation climates in developing countries are, by nature, problematic, characterized by poor business and governance conditions, low educational levels, and mediocre infrastructure'. Essentially, several factors militate against innovations in developing countries, and this will have implications in the adoption of ChatGPT. Mushtaq (2023) states that most people use ChatGPT to create 'content' and that is not using ChatGPT intelligently. However, academics can utilise ChatGPT to create 'structure' for academic writing. Essentially, ChatGPT can be helpful in producing structure for research papers, seminal paper, dissertations, proposals, and theses. Using ChatGPT to produce structure may involve getting an outline of one's research paper for example. The researcher can then use the structure generated by ChatGPT to write to simplify the writing process. This approach can be viewed as consistent with the position on the interaction between the technical and social systems espoused by the socio-technical theory. This perspective views ChatGPT as a tool that can support academics in the same way that Grammarly and Mendeley have been utilised over the years. Professors can also use ChatGPT to review their work as well as students' work. Prompts to ChatGPT to grade students' work have yielded the following response, 'As an AI language model, I do not have the authority or capability to grade assignments for students. However, I can provide feedback on writing and offer suggestions to help improve the quality of written work'. Utilising ChatGPT as a tool to provide feedback on student assessments can speed up the process of grading students' work. In an era of massification of higher education in South Africa, ChatGPT capabilities can greatly ameliorate the pressure that professors have in grading assessments. Other professions on the other hand, have already started infusing ChatGPT in their work. Cheung (2023b) cites a Columbia Judge who used ChatGPT in court to show the growing use of the model around the world. The judge used ChatGPT, 'to ask legal questions and incorporate its answers in the final decision. The goal was to streamline the decision-making process, but the judge was clear that AI does not replace human judgement'. The judge instead of investing hours, days and weeks going through previously cases in the law library, prompted ChatGPT for assistance and thus effectively incorporating ChatGPT. Drawing from this example the author argues that there are intelligent ways of using AI tools such as ChatGPT in the academia. Furthermore, the cited example of the Columbia judge can be utilised to buttress the socio-technical theoretical perspective advocating for complementarity between the technical and social components of the system.

Jobs threatened by ChatGPT in universities

The reality on the ground points to the widespread use of AI in the near future. The future widespread use of AI was acknowledged by Microsoft co-founder Bill Gates in an interview with the CNBC (Goswani, 2023). Bill Gates said in relation to ChatGPT that 'this will change our world' (Goswani, 2023). ChatGPT launched only two months ago, has garnered a remarkable 100 million monthly active users, breaking all previous records in consumer application adoption rates, outpacing even the likes of TikTok and Instagram (Cheung, 2023a). With an average of 13 million unique visitors accessing ChatGPT daily in January, doubling the user base from December, the chatbot's unprecedented success has generated interest from experts in different fields. This growth exhibits the tremendous usage of ChatGPT and the prospects of Google working on introducing a rivalry of ChatGPT shows the demand and growth of AI use globally. Therefore, it is evidently imprudent for academia to invest in building walls against the intrusion of ChatGPT into academic spaces. Cerullo (2023) writes that some jobs are most likely to be replaced by chatbots like ChatGPT. Among the jobs that are likely to be replaced by chatbots are customer service assistants, bookkeeping, and data entry, receptionists, proofreading, retail services, and copywriting among others. While some of these professions are also found in some universities such as receptionists, data entry, service assistants et cetera. Of these jobs, academics would be more interested in the impact of ChatGPT on proofreading. Proofreading is an essential service in academia where student assignments, dissertations, and theses require expert language editing before submission. Academics also make use of proofreading services before submitting their work. However, recent developments in AI will revolutionise proofreading. Universities around the world and in South Africa require

that student dissertations and theses to be language edited by expert language editors. Most universities in South Africa have lists of approved language editors that are recommended to students. Due to demand very often, these language editors are overwhelmed with work from students and professors. Language editing of dissertations and theses often takes time. Additionally, language editing has always been an extra financial burden on the students and professors. However, chatbots can reduce the time needed to language edit a document. Students and professors can now utilise a chatbot without necessarily engaging the services of a language editor. Essentially, ChatGPT brings efficiency to language editing and therefore is likely to replace the work of language editors. Essentially, ChatGPT is likely to render language editors redundant due to its efficiency as a language editor.

South Africa has 12 official languages. Higher education institutions strive to promote these languages through pursuing multilingualism. For example, Stellenbosch University's language policy promotes the use of three languages, English, Afrikaans and IsiXhosa. While the University of Cape Town have implemented initiatives to promote multilingualism such as Multilingualism Education Project (MEP). Essentially, there are several efforts across South African universities to promote multilingualism. These efforts have led to the rise and relevance of translators and developers of electronic glossary. However, ChatGPT offers faster translation services that signals the demise of the need for translators in the future South African universities. Students previously requiring the help of language centres to translate their theses abstract will now use ChatGPT. Additionally, the relevance of glossaries is also under threat as ChatGPT can translate a whole document instead of word-by-word explanations. Thus, this natural language capability of ChatGPT can be viewed as a threat to some jobs in the academia. This threat that ChatGPT poses on some of these jobs can be viewed as an alignment between the technical and social systems within organisation as stated by the socio-technical theory. Jobs that are misaligned will inevitably become redundant such as language editors and translators due to advances in technology. However, these jobs can be aligned with technology by embracing ChatGPT and using it to enhance their work.

University assessments within a ChatGPT environment

The launch of ChatGPT in late 2022 was met with mixed reactions from different professions. Interestingly, the usual front runners with innovations are generally sceptical. Academics have been renowned for being advocates for technological innovations. However, the initial reaction in universities in South Africa to ChatGPT has rather been slow as compared to other professions. Headlines such as 'ChatGPT Applications That Will Blow Your Mind With Their AI Capabilities', 'Can ChatGPT write better SQL than a Data analyst' in other professions have not been replicated in the academia in South Africa. This reaction by other professions is consistent with the central thesis in the socio-technical theory that does not view technology as a threat. Fundamentally, ChatGPT has raised several questions that are informed by the predisposition that profession has towards AI. Professions that have shown an appreciation of the utility of ChatGPT, see opportunities in the use of ChatGPT and are investing in staff development and learning to use ChatGPT. While professions that see ChatGPT as a threat are focused on policing the use of ChatGPT. Essentially, some professions are already seized with positioning themselves in a world with ChatGPT. A world that is waking up to the reality that AI might be the way to go. The academia reaction to ChatGPT is analogous to the reaction to the use of calculators in school classrooms more than half a century ago in the United States of America. Banks (2011, p. 2) states 'some educators feared that students would not be able to retain their knowledge of simple arithmetic if they learned to use a calculator before fully grasping basic mathematical concepts'. However, eventually, 'teachers and curriculum writers searched for ways to teach effectively with the calculator as a pedagogical tool' (Banks, 2011, p. 3). Essentially, universities, and academics need to search for ways of using ChatGPT as a complementary pedagogical tool. The perceived threat of ChatGPT to traditional assessment should spur professors to innovate and improve their assessments. Assessments in mathematics had to change with the introduction of calculators within the classroom. By extension, it can be argued that assessment in universities in South Africa need to be changed in line with the advances in AI such as ChatGPT. However, as a word of caution changing assessments in response to ChatGPT should not entail reverting to traditional and antiquated approaches. Some professors at universities in

South Africa have opted to revert to in-person paper and pen assessment as opposed to online assessments. Essentially, opting to police the students instead of finding ways to assess in a world with AI. Thus, inevitably undoing two decades of strides in integration of Information Communication Technologies.

The perceived threat of ChatGPT on assessment in higher education institution has led some to view it as a bane to assessments. ChatGPT has been conceptualized as a tool that can be used to cheat by students. Headlines in the media of ChatGPT passing exams have further exacerbated the fears that academics have for AI. George Vernadakis reported in the UK Daily Mail that ChatGPT had passed the gold-standard US medical exam (Vernadakis, 2023). While Karen Sloan reported for Reuters states that ChatGPT passed law school exams at the University of Minnesota (Sloan, 2023). Samantha Murphy Kelly reported for the CNN that ChatGPT passed exams from law and business schools (Kelly, 2023). Kelly (2023) wrote 'The powerful new AI chatbot tool recently passed law exams in four courses at the University of Minnesota and another exam at University of Pennsylvania's Wharton School of Business, according to professors at the schools'. These headlines revealing ChatGPT intellectual ability show that the fears of some professors might be justified to a certain extent. Premised on these headlines, justifications for policing the use of ChatGPT have been proffered in academic circles in South Africa and at large. At the epicentre of these arguments are concerted efforts to identify AI generated content due a policing mentality among some academics. However, premised on the reality that AI is here and is here to stay, the paper advocates for ways of enhancing assessments in university in such a way that ChatGPT among other AI tools play a complementary role. Arguments previously offered for the integration of ICT into assessments in universities are still relevant and plausible in the context of using ChatGPT in the academia. Using ICT tools in assessments is not only aimed at improving assessments but also to prepare students for life after examinations, that is the real world. A life that involved working and living in a world that is dominated by ICT use and ultimately AI in future. Therefore, the import of the paper is centred on the argument that assessments should continue to prepare students for the future and that future now involve living and working in a world with AI. Aligning ChatGPT into university assessment is a giant step towards building windmills and harnessing the energy that comes with using ChatGPT. Universities should continue their mandate to prepare students for life after university, life in the world that has embraced ChatGPT. Universities not embracing AI tools would be doing a disservice to the nation and the world that entrusted academia with the responsibility to adequately equip the next generation with adequate skills.

This study has argued for conceptualizing AI as an expansion of universities' technical capacities rather than an existential threat. However, truly harnessing innovation whilst upholding academic integrity requires an alignment between technology and social systems – policies, norms, oversight. This socio-technical imperative emerged as a cross-cutting theme – nearly all literature emphasized responsible assimilation balancing benefits and risks. Effective utilization requires transparency, accountability and defined AI usage scopes complementing human strengths, not outsourcing cognition. Explicitly embedding recommendations within this socio-technical framework strengthens conclusions and policy impact. Specifically, the socio-technical lens underscores that rather than resisting technology, incorporating it into flexible governance frameworks fosters adaptable yet accountable innovation in which disruption strengthens rather than undermines academic excellence. Updating policies, assessments, and training around responsible AI usage enables dynamically harnessing benefits like personalization and efficiency whilst upholding critical analysis and oversight. The conclusions advocate this socio-technical alignment – it is not only an effective analytical framework but the essential, pragmatic mechanism for manifesting this study's vision of AI as a capability expander rather than academic integrity demolisher.

Concluding remarks

This paper has elucidated debates regarding assimilating disruptive innovations like AI, specifically ChatGPT, within academia – analysing anxieties alongside opportunities. Applying socio-technical principles, it was argued that rather than perceiving emerging technologies as threats, they should be viewed as expanding systems' technical capabilities. However, due accountability mechanisms must foster responsible usage complementing human strengths regarding analysis, ethics, and oversight. The study

revealed that whilst representing remarkable technical progress, systems like ChatGPT also introduce uncertainties requiring policy responses. Analysis found that instead of resisting or hastily adopting new technologies, a considered, socially attuned strategy for assimilation is recommended – upholding academic rigor whilst responsibly expanding technical capacities regarding efficiency, personalization and accessibility. Specifically, the paper argued that by conceptualizing tools like ChatGPT as augmenting rather than automating academia's technical subsystem, policies can align innovation with upholding integrity. This entails transparent protocols enabling students and faculty to utilize AI in defined contexts – delegating repetitive tasks whilst retaining human agency over analysis, ideation, and quality control. Rather than perceiving emerging technologies as threats, their responsible assimilation as complements aligned to human oversight through updated policies and pedagogies is advocated.

The study also revealed gaps representing fruitful avenues for future research. Firstly, more empirical evidence is needed regarding ChatGPT's impacts on learning, research, and faculty workloads – longitudinal studies would enrich understanding. Secondly, whilst literature often conceptualizes AI dichotomously as utopian or dystopian, the reality is more complex – nuanced perspectives tracking evolving impacts across contexts over time could better inform policy. Finally, strategies and best practices for transparently yet effectively utilizing AI merit investigation – case studies from innovative institutions would catalyse progress. This paper has argued for a balanced approach towards aligning disruptive innovations with academia's purposes – upholding integrity whilst expanding access and insight. Through updated policies and protocols fostering responsible usage, tools like ChatGPT can play a complementary role – provided oversight remains human. Adopting this socio-technical thinking entails conceptualizing AI as enhancing education's technical capacities, not threatening its social essence – insight over automation, ethics over efficiency. If innovation expands capabilities without undermining critical oversight, academia can responsibly lead in our increasingly high-technology world.

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